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(54) **THERMAL MANAGEMENT SYSTEM**

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(57) **ABSTRACT**

The thermal management system comprises a LT radiator, an oil cooler, a chiller, a nine-way valve and a five-way valve, and an ECU. When heating is required, ECU controls the nine-way valve and the five-way valve when the temperature of the heat transfer medium that is flowing through LT radiator is less than the ambient air temperature, to form a first heat transfer medium circuit in which the oil cooler and LT radiator are connected to the chiller, and when the temperature of the heat transfer medium that is flowing through LT radiator is equal to or higher than the ambient air temperature, controls the nine-way valve and the five-way valve to form a second heat transfer medium circuit in which LT radiator is disconnected from the first heat transfer medium circuit.

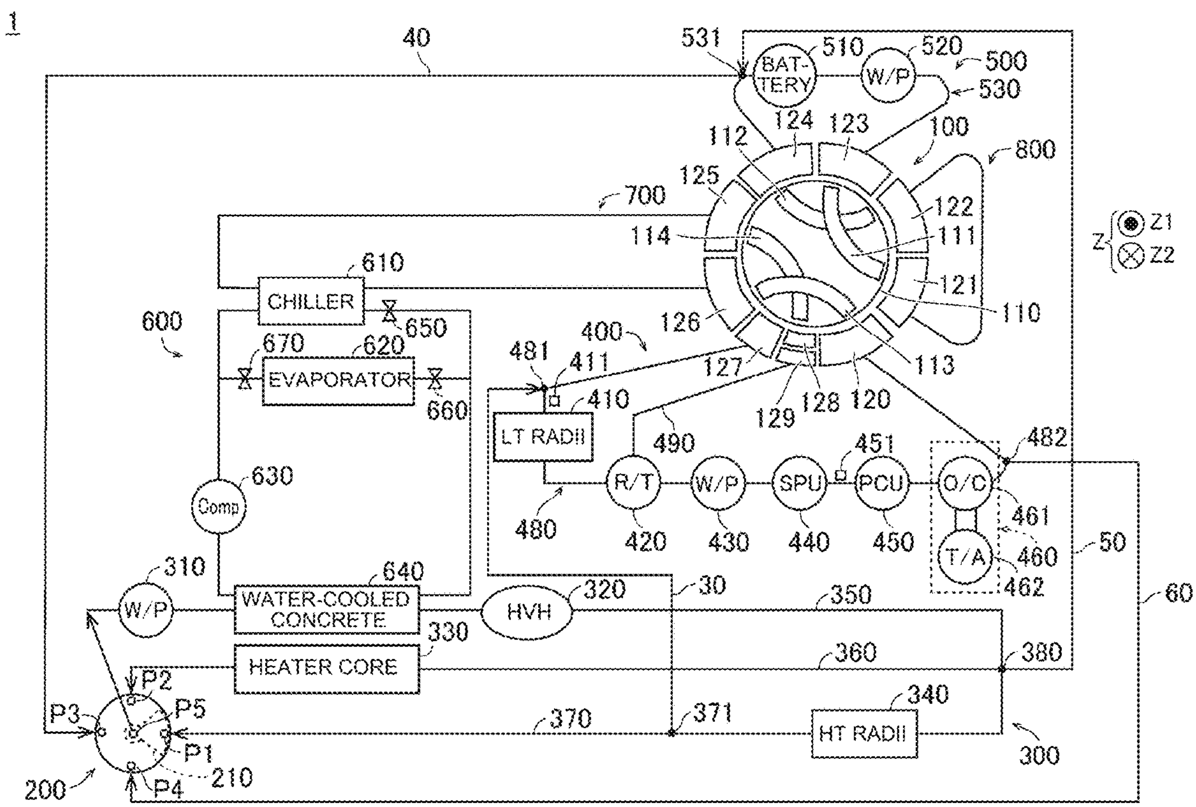


FIG. 1

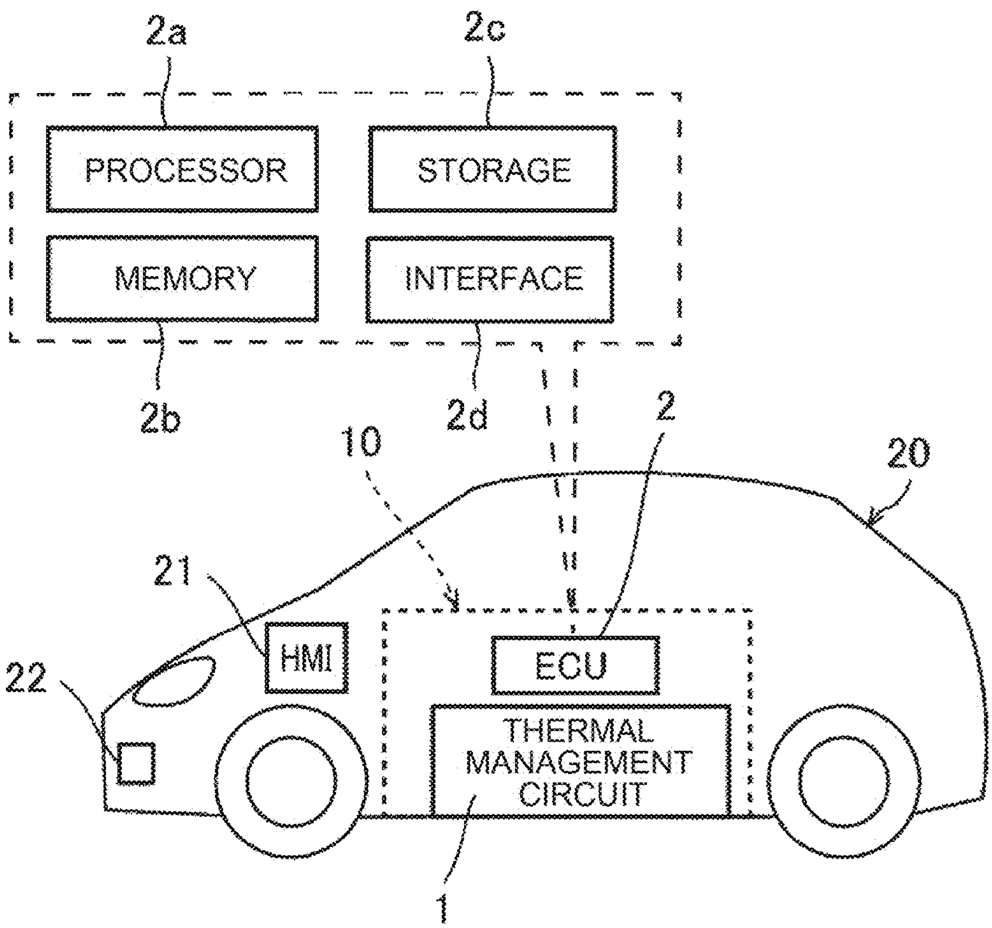

2a
2c
PROCESSOR
STORAGE
MEMORY
INTERFACE
2b
2d
10
2
20
21
HMI
22
ECU
THERMAL
MANAGEMENT
CIRCUIT
1

FIG. 2

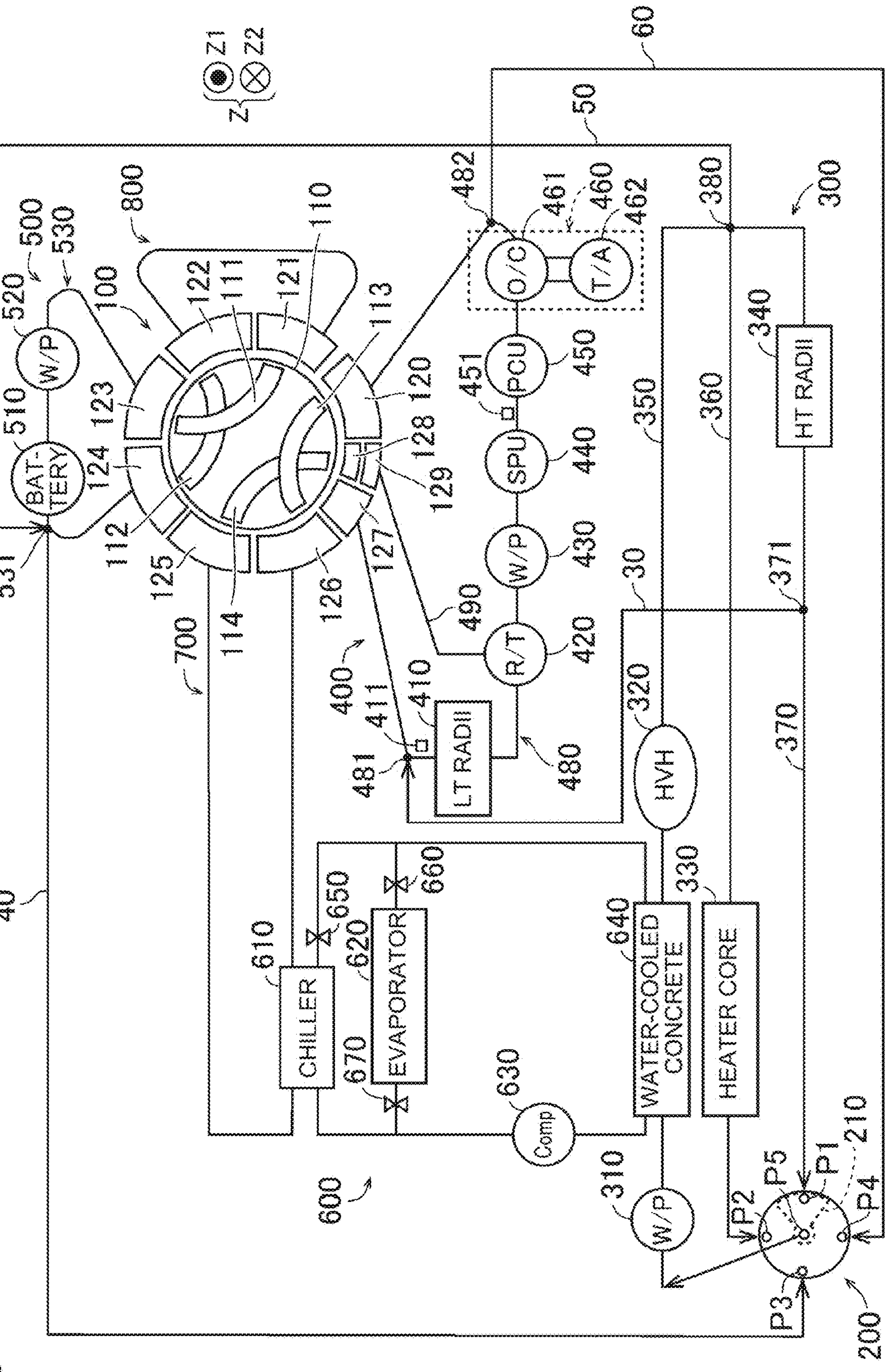

1
40
50
60
30
Z
Z1
Z2
BAT-TERY
510
W/P
520
530
500
531
100
800
110
111
112
113
114
120
121
122
123
124
125
126
127
128
129
482
O/C
461
T/A
462
460
PCU
450
451
SPU
440
W/P
430
R/T
420
490
400
LT RADII
410
411
481
480
700
380
300
HT RADII
340
350
360
371
370
HVH
320
330
HEATER CORE
WATER-COOLED CONCRETE
640
CHILLER
610
EVAPORATOR
620
650
660
670
Comp
630
600
W/P
310
P1
P2
P3
P4
P5
210
200

FIG. 3

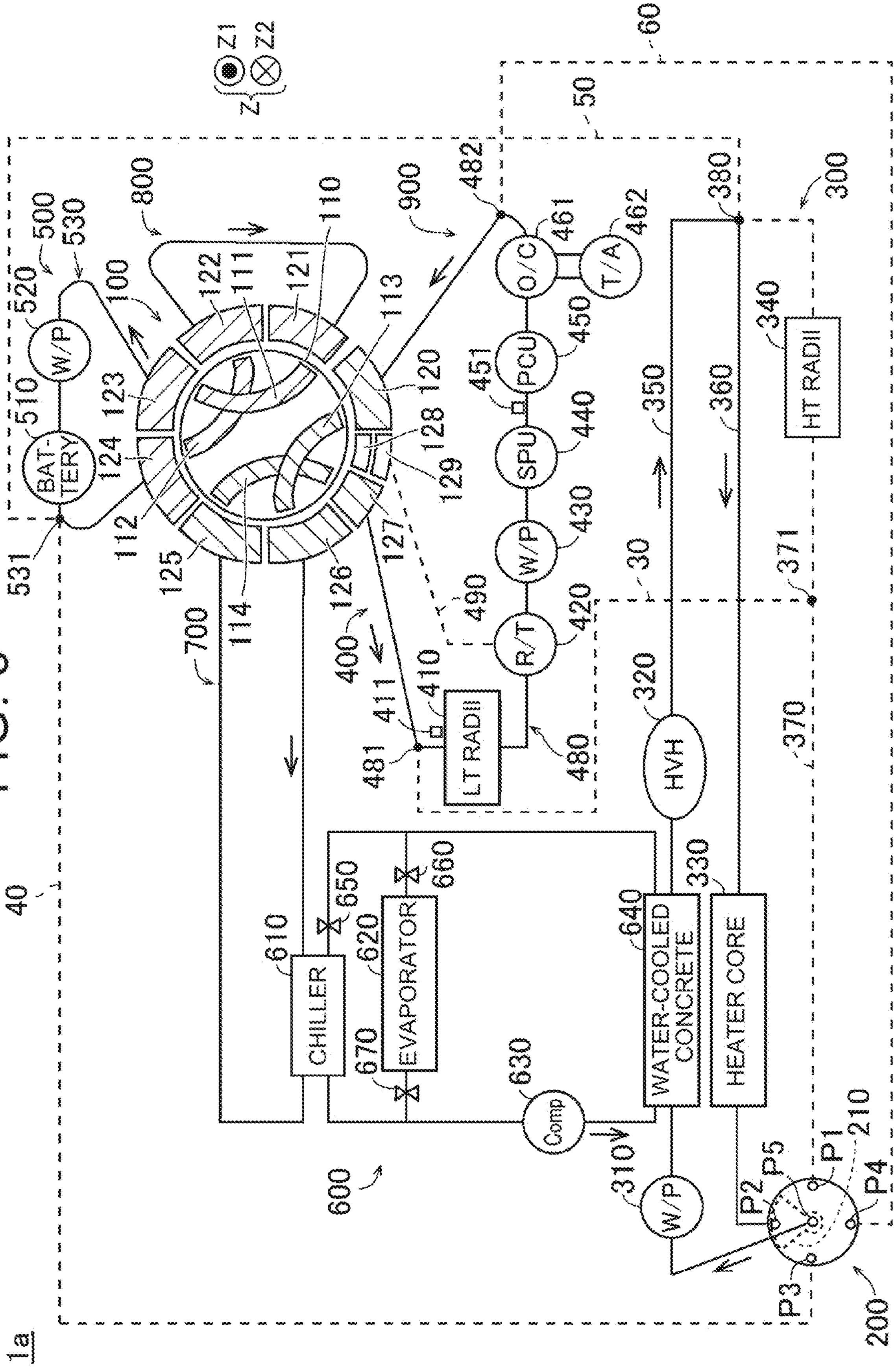
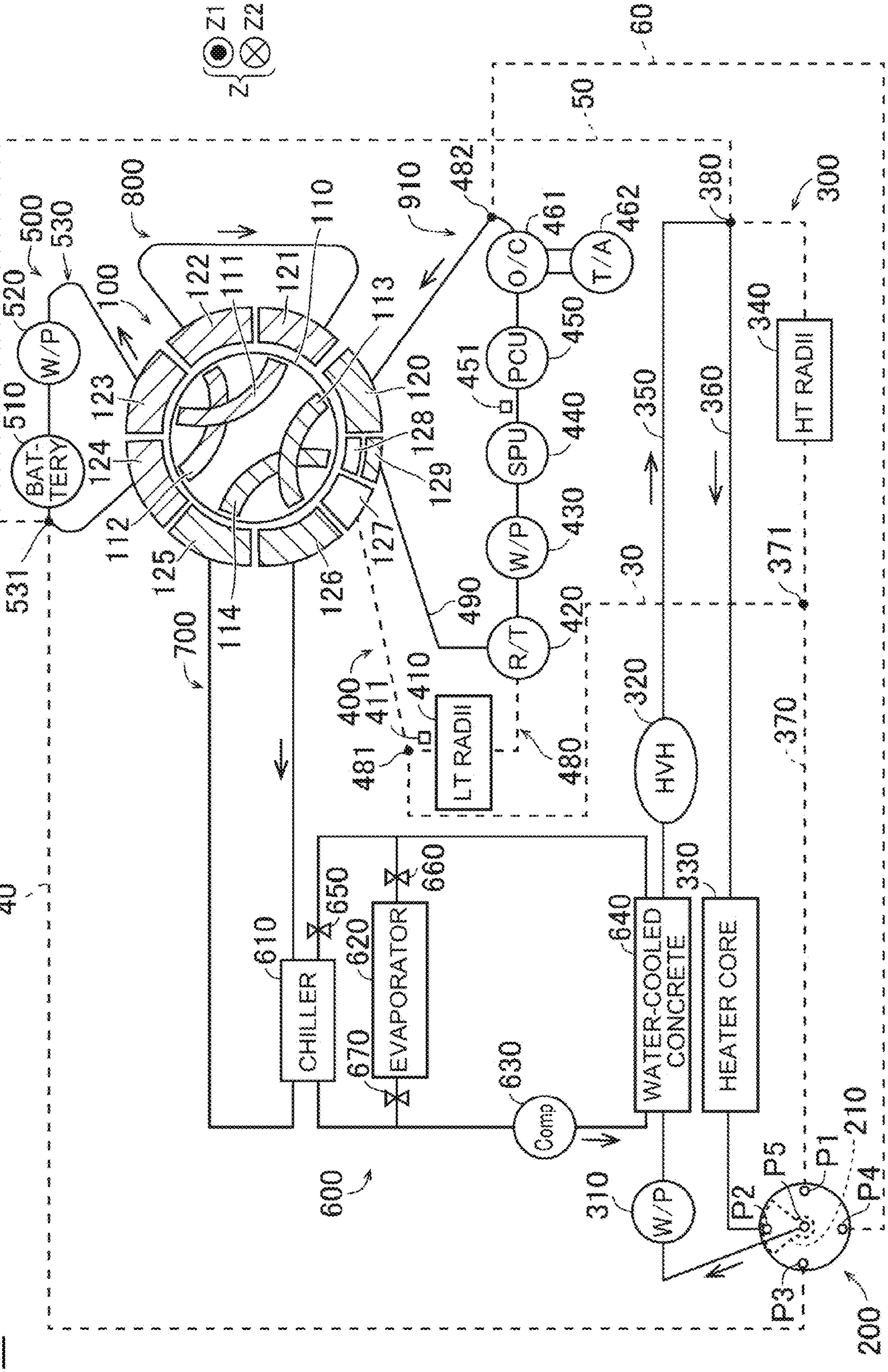

1a
40
50
60
30
Z
Z1
Z2
BAT-TERY
W/P
LT RADII
HT RADII
R/T
SPU
PCU
O/C
T/A
HVH
CHILLER
EVAPORATOR
WATER-COOLED CONCRETE
HEATER CORE
Comp
P1
P2
P3
P4
P5
100
110
111
112
113
114
120
121
122
123
124
125
126
127
128
129
200
210
300
310
320
330
340
350
360
370
371
380
400
410
411
420
430
440
450
451
461
462
480
481
482
490
500
510
520
530
531
600
610
620
630
640
650
660
670
700
800
900

FIG. 4



BAT-TERY
W/P
CHILLER
EVAPORATOR
Comp
WATER-COOLED CONCRETE
HEATER CORE
HVH
LT RADII
HT RADII
R/T
SPU
PCU
O/C
T/A
P1
P2
P3
P4
P5
Z
Z1
Z2

1b

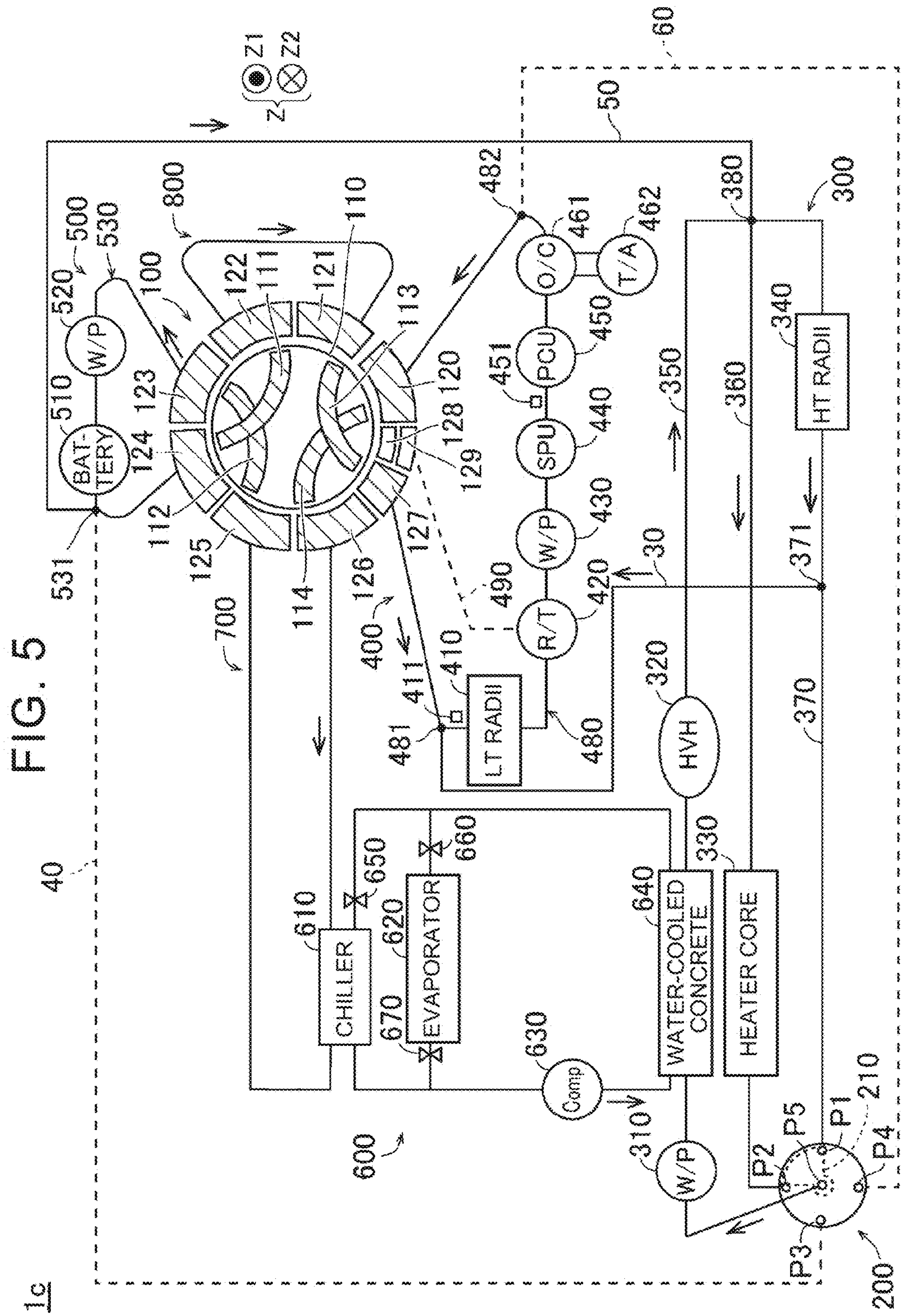
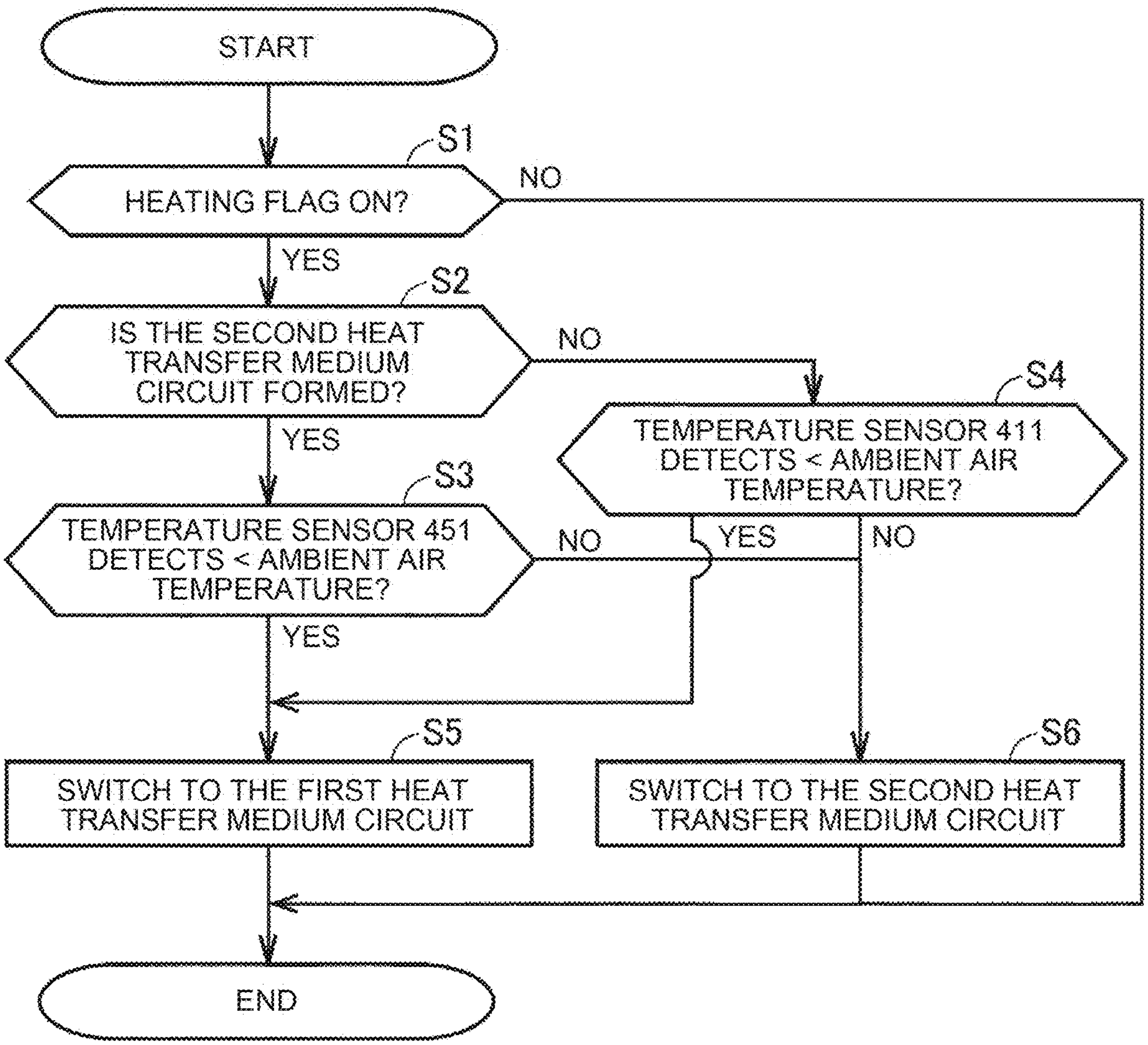

1c
40
50
60
Z
Z1
Z2
BAT-
TERY
W/P
500
510
520
530
531
100
800
110
111
112
113
114
120
121
122
123
124
125
126
127
128
129
482
O/C
T/A
461
462
PCU
SPU
W/P
R/T
451
450
440
430
420
490
400
410
411
481
480
LT RADII
700
30
HVH
320
350
360
380
300
340
HT RADII
370
371
CHILLER
EVAPORATOR
610
620
650
660
670
Comp
630
WATER-COOLED
CONCRETE
640
HEATER CORE
330
600
W/P
310
P1
P2
P3
P4
P5
210
200

FIG. 5

FIG. 6


START
S1
HEATING FLAG ON?
NO
YES
S2
IS THE SECOND HEAT TRANSFER MEDIUM CIRCUIT FORMED?
NO
YES
S4
TEMPERATURE SENSOR 411 DETECTS < AMBIENT AIR TEMPERATURE?
S3
TEMPERATURE SENSOR 451 DETECTS < AMBIENT AIR TEMPERATURE?
YES
NO
NO
YES
S5
SWITCH TO THE FIRST HEAT TRANSFER MEDIUM CIRCUIT
S6
SWITCH TO THE SECOND HEAT TRANSFER MEDIUM CIRCUIT
END

THERMAL MANAGEMENT SYSTEM

CROSS-REFERENCE TO RELATED APPLICATION

[0001] This application claims priority to Japanese Patent Application No. 2024-037233 filed on Mar. 11, 2024, incorporated herein by reference in its entirety.

BACKGROUND

1. Technical Field

[0002] The present disclosure relates to a thermal management system.

2. Description of Related Art

[0003] Japanese Unexamined Patent Application Publication No. 2020-185829 (JP 2020-185829 A) discloses an electrified vehicle that is equipped with an in-vehicle temperature control device including a low-temperature radiator and a power control unit (PCU) heat exchanger. The low-temperature radiator and the PCU heat exchanger are connected in series with a chiller when heating is requested in the electrified vehicle. In this case, heat that is supplied from a PCU to a coolant through the PCU heat exchanger, and heat that is supplied to the coolant from the ambient air in the low-temperature radiator (heat absorption by the low-temperature radiator) are used for heating.

SUMMARY

[0004] However, in JP 2020-185829 A, when temperature of the coolant flowing through the low-temperature radiator is higher than ambient air temperature, for example, heat will be undesirably dissipated from the coolant to the ambient air in the low-temperature radiator. Thus, there is a concern that heating efficiency will deteriorate.

[0005] The present disclosure has been made to solve the above problems, and an object thereof is to provide a thermal management system capable of suppressing decrease in heating efficiency due to the ambient air temperature being high.

[0006] An aspect of the present disclosure provides a thermal management system in which a heat transfer medium circulates, the thermal management system including a radiator, a drive device that includes a heat exchanger and that is configured to generate a driving force, a chiller device, a switching device that switches a flow path of the heat transfer medium, and a control device that controls the switching device.

[0007] When heating is requested, the control device may control the switching device to establish a first heat transfer medium circuit in which the heat exchanger, the radiator, and the chiller device are connected, when a temperature of the heat transfer medium that is flowing through the radiator is lower than ambient air temperature, and control the switching device to establish a second heat transfer medium circuit in which the radiator is isolated from the first heat transfer medium circuit, when the temperature of the heat transfer medium that is flowing through the radiator is no lower than the ambient air temperature.

[0008] In the thermal management system according to an aspect of the present disclosure, as described above, the first heat transfer medium circuit is established when the temperature of the heat transfer medium that is flowing through the radiator is lower than the ambient air temperature, when heating is requested. Further, the second heat transfer medium circuit is established when the temperature of the heat transfer medium that is flowing through the radiator is no lower than the ambient air temperature. Accordingly, when heat is supplied from the ambient air to the heat transfer medium through the radiator due to the temperature of the heat transfer medium that is flowing through the radiator being lower than the ambient air temperature, the heat supplied to the heat transfer medium in both the radiator and the heat exchanger can be used for heating. Also, when heat is released from the heat transfer medium to the ambient air through the radiator due to the temperature of the heat transfer medium that is flowing through the radiator being no lower than the ambient air temperature, only the heat supplied to the heat transfer medium in the heat exchanger can be used for heating. In this case, the radiator does not contribute to heating, and accordingly decrease in heating efficiency due to heat dissipation from the radiator can be suppressed. Thus, decrease in heating efficiency due to the ambient air temperature being high can be suppressed.

[0009] The first heat transfer medium circuit may include a first series circuit in which the heat exchanger, the radiator, and the chiller device are connected in series. The second heat transfer medium circuit may include a second series circuit in which the radiator is isolated from the first series circuit and the heat exchanger and the chiller device are connected in series.

[0010] Such a configuration enables easy switching between whether the radiator contributes to heating, by switching between the first series circuit and the second series circuit.

[0011] The thermal management system may further include a first temperature sensor that detects the temperature of the heat transfer medium that is flowing into the radiator.

[0012] When heating is requested, the control device may control the switching device to establish the first heat transfer medium circuit, when a detected value of the first temperature sensor is lower than the ambient air temperature, and control the switching device to establish the second heat transfer medium circuit, when the detected value of the first temperature sensor is no lower than the ambient air temperature. Such a configuration enables the switching device to be controlled based on a relation between the temperature of the heat transfer medium that is flowing into the radiator and the ambient air temperature, and accordingly the switching device can be controlled based on the relation between the heat transfer medium prior to heat exchange in the radiator and the ambient air temperature. Thus, when there is a temperature condition in which heat will be dissipated to the ambient air at the radiator (i.e., temperature of heat transfer medium >ambient air temperature), the first heat transfer medium circuit can be suppressed from being established more reliably.

[0013] The switching device may include a nine-way valve and a five-way valve.

[0014] The control device may switch between the first heat transfer medium circuit and the second heat transfer medium circuit by controlling each of the nine-way valve and the five-way valve.

[0015] Such a configuration enables easy switching between the first heat transfer medium circuit and the second heat transfer medium circuit by switching the flow path of the heat transfer medium using two different multi-way valves (nine-way valve and five-way valve).

[0016] A second temperature sensor that detects the temperature of the heat transfer medium that is flowing through the heat exchanger may be provided.

[0017] The control device may control the switching device to establish the first heat transfer medium circuit when a detected value of the second temperature sensor becomes lower than the ambient air temperature in a state in which the second heat transfer medium circuit is established.

[0018] With such a configuration, even when the heat transfer medium does not flow through the radiator, switching can be appropriately performed, from the second heat transfer medium circuit to the first heat transfer medium circuit, based on the detected value of the second temperature sensor.

[0019] According to the present disclosure, decrease in heating efficiency due to a high ambient air temperature can be suppressed.

BRIEF DESCRIPTION OF THE DRAWINGS

[0020] Features, advantages, and technical and industrial significance of exemplary embodiments of the disclosure will be described below with reference to the accompanying drawings, in which like signs denote like elements, and wherein:

[0021] FIG. **1** is a diagram illustrating a configuration of a vehicle equipped with a thermal management system according to an embodiment;

[0022] FIG. **2** is a diagram illustrating a configuration of a thermal management circuit of a thermal management system according to an embodiment;

[0023] FIG. **3** is a diagram illustrating a first heat transfer medium circuit of a thermal management circuit according to an embodiment;

[0024] FIG. **4** is a diagram illustrating a second heat transfer medium circuit of a thermal management circuit according to an embodiment;

[0025] FIG. **5** is a diagram illustrating a third heat transfer medium circuit of a thermal management circuit according to an embodiment; and

[0026] FIG. **6** is a diagram illustrating a control flowchart of an ECU according to an embodiment.

DETAILED DESCRIPTION OF EMBODIMENTS

[0027] Hereinafter, an embodiment of the present disclosure will be described in detail with reference to the drawings. The same or corresponding parts in the drawings are designated by the same reference characters and repetitive description will be omitted.

[0028] As illustrated in FIG. **1**, a configuration in which the thermal management system **10** according to the present disclosure is mounted on a electrified vehicle **20** will be described below. Electrified vehicle **20** is preferably a vehicle equipped with a battery **510** for traveling (described below, FIG. **2**). Electrified vehicle **20** is, for example, battery electric vehicle (BEV: Battery Electric Vehicle). Also, electrified vehicle **20** is, for example, hybrid electric vehicle (HEV: Hybrid Electric Vehicle), plug-in hybrid electric vehicle (PHEV: Plug-in Hybrid Electric Vehicle), or fuel cell electric vehicle (FCEV: Fuel Cell Electric Vehicle). However, the use of the thermal management system according to the present disclosure is not limited to a vehicle.

[0029] Electrified vehicle **20** includes a thermal management system **10**, an HMI (Human Machine Interface) **21**, and an ambient air temperature sensor **22**. HMI **21** includes, for example, a car navigation device and the like. The ambient air temperature sensor **22** detects the ambient air temperature.

[0030] The thermal management system **10** includes a thermal management circuit **1** and an ECU (Electronic Control Unit) **2**. Note that ECU **2** is an exemplary "control device" of the present disclosure.

[0031] ECU **2** includes a processor **2***a*, a memory **2***b*, a storage **2***c*, and an interface **2***d*.

[0032] The processor **2***a* is, for example, CPU (Central Processing Unit) or MPU (Micro-Processing Unit). The memory **2***b* is, for example, RAM (Random Access Memory). The storage **2***c* is a rewritable non-volatile memory such as HDD (Hard Disk Drive), SSD (Solid State Drive), and flash memory. The storage **2***c* stores a system program including OS (Operating System) and a control program including computer-readable code required for control operations. The processor **2***a* realizes various processes by reading out a system program and a control program, expanding the program into a memory **2***b*, and executing the program. The interface **2***d* controls the communication between ECU **2** and the components of the thermal management circuit **1**.

[0033] ECU **2** generates a control command based on a sensor value (for example, a detected value of the ambient air temperature sensor **22**) acquired from various sensors included in the thermal management circuit **1**, a user operation (for example, an operation requesting heating) received by HMI **21**, and the like. ECU **2** outputs the generated control command to the thermal management circuit **1**. ECU **2** may be divided into a plurality of ECU for each function. Further, although FIG. **1** shows an exemplary ECU **2** including one processor **2***a*, ECU **2** may include a plurality of processors. The same applies to the memory **2***b* and the storage **2***c*.

Configuration of the Thermal Management Circuit

[0034] FIG. **2** is a diagram illustrating an example of a configuration of the thermal management circuit **1** according to the present embodiment. The thermal management circuit **1** includes a nine-way valve **100**, a five-way valve **200**, a high-temperature circuit **300**, a unit circuit **400**, a battery circuit **500**, a refrigeration cycle **600**, a flow path **700**, and a flow path **800**. The thermal management circuit **1** includes a flow path **30**, a flow path **40**, a flow path **50**, and a flow path **60**. Note that each of the nine-way valve **100** and the five-way valve **200** is an example of a "switching device" of the present disclosure.

[0035] The nine-way valve **100** includes a valve body **110** and outer sections **120** to **129**. The valve body **110** has a cylindrical shape extending in the Z direction, and is configured to be rotatable about a central axis line (not shown) of the valve body **110**. The valve body **110** is surrounded by the outer sections **120** to **129** when viewed from **Z1**. The valve body **110** rotates in accordance with a control command from ECU **2** (FIG. **1**).

[0036] The valve body **110** is provided with an internal flow path **111**, an internal flow path **112**, an internal flow path **113**, and an internal flow path **114**. When the valve body **110** rotates in accordance with a control command from ECU **2**, the positions of the internal flow paths **111** to

114 are changed. As a result, the connection (connection combination) between the outer sections 120 to 129 and the internal flow paths 111 to 114 is changed.

[0037] Details will be described later.

[0038] The internal flow path 111 is disposed Z1 the internal flow path 112. The internal flow path 113 is disposed Z1 the internal flow path 114. The internal flow path 111 and the internal flow path 113 are arranged at the same position in the Z direction. The internal flow path 112 and the internal flow path 114 are arranged at the same position in the Z direction.

[0039] The outer sections 120 to 129 are circumferentially arranged on the outer peripheral side of the valve body 110. The outer sections 120 to 129 are isolated from each other. Although the outer sections 128 and 129 are shown as being arranged side by side in FIG. 2 for clarity, they are actually stacked in the Z direction. The outer section 129 is disposed Z2 the outer section 128.

[0040] Each of the outer sections 120 to 126 has substantially the same width in the circumferential direction of the valve body 110. The circumferential width of each of the outer sections 127 to 129 is less than the circumferential width of the outer sections 120 to 126.

[0041] The five-way valve 200 has a cylindrical shape extending in the Z direction. The five-way valve 200 has a P5 from the port P1. P4 from the port P1 is an inlet port that allows the heat transfer medium to flow into the five-way valve 200. The port P5 is an outlet port that allows the heat transfer medium to flow out of the five-way valve 200. P4 from the port P1 is in communication with a lower compartment (not shown) that is isolated from one another. That is, the five-way valve 200 is provided with four lower compartments that are isolated from each other. Although not shown, the four sections have the same shape (a sector shape having a central angle of 90 degrees when viewed from Z1 side). The port P5 communicates with an upper compartment (not shown) provided Z1 of the four lower compartments.

[0042] The flow condition of the heat transfer medium from the port P1 to P5 is controlled by a control command from ECU 2 (FIG. 1). Specifically, ECU 2 changes the position of the opening 210. Inside the five-way valve 200, a partition plate (not shown) for partitioning the four lower and upper partitions is provided, the opening 210 is formed in the partition plate. The partition plate is rotationally moved about the central axial line of the five-way valve 200 having a cylindrical shape by a control command from ECU 2. As a result, the lower section overlapping with the opening 210 in the Z direction changes. The port P5 is only in communication with a port that communicates with the lower compartment that overlaps the opening 210 in the Z-direction.

[0043] The opening 210 has a sector shape with a central angle of about 90 degrees when viewed from Z1. Thus, one or two of the four lower compartments may simultaneously overlap the opening 210 in the Z direction. This causes one or two ports of P4 from the port P1 to communicate with the port P5.

[0044] The high temperature circuit 300 includes a water pump 310, a HVH (High Voltage Heater) 320, a heater core 330, and a HT (High Temperature) radiator 340. The water pump 310 circulates the heat transfer medium in the high temperature circuit 300 in accordance with a control command from ECU 2 (FIG. 1).

[0045] The high temperature circuit 300 includes a flow path 350, a flow path 360, and a flow path 370. The flow path 350 connects the port P5 and the branch point 380. The flow path 350 is provided with a water pump 310 and a HVH 320. The flow path of the flow path 350 between the water pump 310 and HVH 320 is also connected to a water-cooled condenser 640 of the refrigeration cycle 600, which will be described later. That is, in the water-cooled condenser 640, heat exchange is performed between the heat transfer medium circulating in the refrigeration cycle 600 and the heat transfer medium that is flowing through the flow path 350.

[0046] The flow path 360 connects the branch point 380 and the port P2. The flow path 360 is provided with a heater core 330.

[0047] The flow path 370 connects the branch point 380 and the port P1. A HT radiator 340 is provided in the flow path 370.

[0048] Unit circuit 400 includes a LT (Low Temperature) radiator 410, a reserve tank 420, and a water pump 430. The unit circuit 400 includes a SPU (Smart Power Unit) 440, a PCU (Power Control Unit) 450, and a drive unit 460. The drive unit 460 is a device capable of generating a driving force of electrified vehicle 20 (FIG. 1). The drive unit 460 includes an oil cooler (O/C) 461 and a transaxle (T/A) 462. Note that LT radiator 410 and the oil cooler 461 are exemplary "radiators" and "heat exchangers" of the present disclosure, respectively. Each of PCU 450 and the drive unit 460 is an exemplary "drive device" of the present disclosure.

[0049] The water pump 430 circulates the heat transfer medium in accordance with a control command from ECU 2 (FIG. 1). The water pump 430 delivers a heat transfer medium to SPU 413.

[0050] The unit circuit 400 includes a temperature sensor 411 and a temperature sensor 451. The temperature sensor 411 detects the temperature of the heat transfer medium that is flowing through LT radiator 410. Specifically, the temperature sensor 411 detects the temperature of the heat transfer medium that is flowing into LT radiator 410. For example, the temperature sensor 411 detects the temperature of the heat transfer medium passing through the heat transfer medium inlet of LT radiator 410.

[0051] The temperature sensor 451 detects the temperature of the heat transfer medium that is flowing through PCU 450 (the heat exchanger of PCU 450). The temperature sensor 451 may detect the temperature of the heat transfer medium immediately before flowing through PCU 450 (the heat exchanger of PCU 450) (for example, the heat transfer medium inlet in the heat exchanger of PCU 450). Further, the temperature sensor 451 may detect the temperature of the heat transfer medium that is flowing through the oil cooler 461 or the heat transfer medium immediately before flowing through the oil cooler 461 (for example, the heat transfer medium inlet in the oil cooler 461). The heat exchanger of PCU 450 is an exemplary "heat exchanger" of the present disclosure.

[0052] The unit circuit 400 includes a flow path 480 and a flow path 490. The flow path 480 connects the outer section 127 and the outer section 120 of the nine-way valve 100. The flow path 480 is provided with a LT radiator 410, a reserve tank 420, a water pump 430, a SPU 440, PCU 450, and an oil cooler 461. SPU 440, like PCU 450, includes a heat exchanger (not shown) that exchanges heat with the heat transfer medium of the flow path 480. The oil cooler

461 exchanges heat between the transaxle **462** and the heat transfer medium in the flow path **480**. Instead of the oil cooler **461**, a transaxle **462** may be provided in the flow path **480**.

[0053] The flow path **490** connects the outer section **129** of the nine-way valve **100** and the reserve tank **420**.

[0054] The battery circuit **500** includes a battery **510**, a water pump **520**, and a flow path **530**. The water pump **520** delivers a heat transfer medium to the battery **510** side. The water pump **520** circulates the heat transfer medium in accordance with a control command from ECU **2** (FIG. **1**). Note that the battery **510** is an example of a "power storage device" of the present disclosure.

[0055] The flow path **530** connects the outer section **123** and the outer section **124** of the nine-way valve **100**. The flow path **530** is provided with a battery **510** and a water pump **520**. The battery **510** includes a heat exchanger (not shown) that exchanges heat with the heat transfer medium of the flow path **530**.

[0056] The refrigeration cycle **600** includes a chiller **610**, an evaporator **620**, a compressor **630**, and a water-cooled condenser **640**. The refrigeration cycle **600** includes an expansion valve **650**, an expansion valve **660**, and a EPR (Evaporative Pressure Regulator) **670**. A heat transfer medium (a gas-phase refrigerant or a liquid-phase refrigerant) circulating in the refrigeration cycle **600** flows through one/both of the first path and the second path. The first path is the path of compressor **630**-water cooled condenser **640**-expansion valve **660**-evaporator **620**-EPR **670**-compressor **630**. The second path is the path of compressor **630**-water cooled condenser **640**-expansion valve **650**-chiller **610**-compressor **630**. The chiller **610** is an example of a "chiller device" of the present disclosure.

[0057] The flow path **700** connects the outer section **125** and the outer section **126** of the nine-way valve **100**. The flow path **700** is also connected to the chiller **610** of the refrigeration cycle **600**. That is, the heat transfer medium that is flowing through the flow path **700** and the heat transfer medium of the refrigeration cycle **600** are heat-exchanged in the chiller **610**.

[0058] The flow path **800** connects the outer section **121** and the outer section **122** of the nine-way valve **100**. A device or the like is not provided in the flow path **800**.

[0059] The flow path **30** connects the branch point **371** and the merging portion **481**. The branch point **371** is provided between HT radiator **340** and the five-way valve **200** in the flow path **370** of the high temperature circuit **300**. The merging portion **481** is provided between LT radiator **410** and the nine-way valve **100** in the flow path **480** of the unit circuit **400**. That is, the heat transfer medium of the flow path **370** branched into the flow path **30** at the branch point **371** is merged into the heat transfer medium of the flow path **480** at the merging portion **481**.

[0060] The flow path **40** connects the port P**3** of the nine-way valve **100** and a branch point **531** of the flow path **530** of the battery circuit **500** provided between the battery **510** and the outer section **124**. That is, the heat transfer medium in the flow path **530** branched into the flow path **40** at the branch point **531** flows into the five-way valve **200** from the port P**3**.

[0061] The flow path **50** connects the branch point **380** of the high-temperature circuit **300** and the branch point **531** of the battery circuit **500**. That is, the heat transfer medium of the high-temperature circuit **300** (the flow path **350**) branched into the flow path **50** at the branch point **380** merges into the flow path **530** at the branch point **531**.

[0062] The flow path **60** connects the port P**4** of the five-way valve **200** and a branch point **482** of the flow path **480** of the unit circuit **400** between the drive unit **460** (oil cooler **461**) and the outer section **120** of the nine-way valve **100**. That is, the heat transfer medium in the flow path **480** branched into the flow path **60** at the branch point **482** flows into the five-way valve **200** from the port P**4**.

[0063] When the temperature of the heat transfer medium that is flowing through LT radiator is higher than the ambient air temperature, heat is dissipated from the heat transfer medium to the ambient air in LT radiator in a conventional thermal management system. Thus, there is a concern that heating efficiency will deteriorate.

[0064] Therefore, in the present embodiment, ECU **2** controls the nine-way valve **100** and the five-way valve **200** when the temperature of the heat transfer medium that is flowing through LT radiator **410** is lower than the ambient air temperature when heating is requested. This forms a first heat transfer medium circuit **1***a* (FIG. **3**) in which each of the oil cooler **461** and LT radiator **410** is connected to the chiller **610**. In addition, ECU **2** controls the nine-way valve **100** and the five-way valve **200** when the temperature of the heat transfer medium that is flowing through LT radiator is equal to or higher than the ambient air temperature when heating is requested. Accordingly, the second heat transfer medium circuit **1***b* (FIG. **4**) in which LT radiator **410** is disconnected from the first heat transfer medium circuit **1***a* is formed.

First Heat Transfer Medium Circuit

[0065] FIG. **3** is a diagram illustrating a first heat transfer medium circuit **1***a* of the thermal management circuit **1**. In the first heat transfer medium circuit **1***a*, LT radiator **410**, the oil cooler **461** (drive unit **460** (FIG. **2**)), and the chiller **610** are provided in a common circulation circuit. Discuss in detail below.

[0066] The internal flow path **111** of the nine-way valve **100** connects the outer section **121** and the outer section **123**. The internal flow path **112** connects the outer section **122** and the outer section **124**. The internal flow path **113** connects the outer section **120** and the outer section **126**. The internal flow path **114** connects the outer section **125** and the outer section **127**.

[0067] In the five-way valve **200**, the opening **210** overlaps only the lower compartment that communicates with the port P**2**. Thus, the port P**5** is in communication with only the port P**2** of P**4** from the port P**1**.

[0068] As a result, the heat transfer medium circulates through the circuit of the five-way valve **200** (port P**5**)-the water pump **310**-the water-cooled condenser **640**-HVH **320**-heater core **330**-the five-way valve **200** (port P**2**) (the circulation circuit in the high-temperature circuit **300**).

[0069] The heat transfer medium also circulates in the circuit of reserve tank **420**-water pump **430**-SPU **440**-PCU **450**-oil cooler **461**-outer compartment **120**-internal flow path **113**-outer compartment **126**-chiller **610**-outer compartment **125**-internal flow path **114**-outer compartment **127**-LT radiator **410**-reserve tank **420**.

[0070] That is, in the first heat transfer medium circuit **1***a*, a series circuit **900** is formed in which LT radiator **410**, the oil cooler **461** (and PCU **450** heat exchanger), and the chiller **610** are connected in series. The series circuit **900** includes a portion of the flow path **480**, an internal flow path **113**, and a portion of the flow path **700**. The series circuit **900** is an example of a "first series circuit" of the present disclosure.

[0071] In addition, the heat transfer medium circulates in the circuits of the water pump **520**, the battery **510**, the outer section **124**, the internal flow path **112**, the outer section **122**, the flow path **800**, the outer section **121**, the internal flow path **111**, the outer section **123**, and the water pump **520**.

[0072] In addition, in the refrigeration cycle **600**, the heat transfer medium circulates through the circuit of the chiller **610**-compressor **630**-water-cooled condenser **640**-expansion valve **650**-chiller **610**.

[0073] In the first heat transfer medium circuit **1***a*, heat exchange with the ambient air is performed in LT radiator **410**, and heat exchange is performed in the oil cooler **461** (and the heat exchanger of PCU **450**), and the heat transfer medium flows through the chiller **610**. The heat supplied to the refrigeration cycle **600** through the chiller **610** is used for heating by the heater core **330** in the high-temperature circuit **300**.

[0074] In the first heat transfer medium circuit **1***a*, the heat transfer medium does not flow through each of the flow path **30**, the flow path **40**, the flow path **50**, and the flow path **60**. In the first heat transfer medium circuit **1***a*, the heat transfer medium does not flow through the flow path **490**.

Second Heat Transfer Medium Circuit

[0075] FIG. **4** is a diagram illustrating a second heat transfer medium circuit **1***b* of the thermal management circuit **1**. The second heat transfer medium circuit **1***b* is a circuit in which LT radiator **410** is disconnected from the first heat transfer medium circuit **1***a*.

[0076] The internal flow path **111** of the nine-way valve **100** connects the outer section **121** and the outer section **123**. The internal flow path **112** connects the outer section **122** and the outer section **124**. The internal flow path **113** connects the outer section **120** and the outer section **126**. The internal flow path **114** connects the outer section **125** and the outer section **129**. That is, only the outer sections connected by the internal flow path **114** are different from the first heat transfer medium circuit **1***a*.

[0077] Similar to the first heat transfer medium circuit **1***a*, in the five-way valve **200**, the opening **210** overlaps only the lower compartment that communicates with the port P2. Thus, similar to the first heat transfer medium circuit **1***a*, the port P**5** is in communication with only the port P**2** of P**4** from the port P**1**.

[0078] As a result, similar to the first heat transfer medium circuit **1***a*, the heat transfer medium circulates through the circuit of the five-way valve **200** (port P**5**)-water pump **310**-water cooled condenser **640**-HVH **320**-heater core **330**-five-way valve **200** (port P**2**).

[0079] The heat transfer medium also circulates in the circuit of the reserve tank **420**-water pump **430**-SPU **440**-PCU **450**-oil cooler **461**-outer compartment **120**-internal flow path **113**-outer compartment **126**-chiller **610**-outer compartment **125**-internal flow path **114**-outer compartment **129**-reserve tank **420**. That is, LT radiator **410** is bypassed with respect to the circulation circuit in the first heat transfer medium circuit **1***a* (FIG. **3**).

[0080] Specifically, in the second heat transfer medium circuit **1***b*, a series circuit **910** in which LT radiator **410** is disconnected from the series circuit **900** (FIG. **3**) is formed. In the series circuit **910**, the oil cooler **461** (and PCU **450** heat exchanger) and the chiller **610** are connected in series. The series circuit **910** includes a portion of the flow path **480** downstream of LT radiator **410**, an internal flow path **113**, and a portion of the flow path **700**. The series circuit **910** is an example of a "second series circuit" of the present disclosure.

[0081] In the second heat transfer medium circuit **1***b*, similarly to the first heat transfer medium circuit **1***a*, the heat transfer medium circulates in the circuits of the water pump **520**, the battery **510**, the outer section **124**, the internal flow path **112**, the outer section **122**, the flow path **800**, the outer section **121**, the internal flow path **111**, the outer section **123**, and the water pump **520**.

[0082] In the second heat transfer medium circuit **1***b*, similarly to the first heat transfer medium circuit **1***a*, in the refrigeration cycle **600**, the heat transfer medium circulates in the circuit of the chiller **610**-compressor **630**-water-cooled condenser **640**-expansion valve **650**-chiller **610**.

[0083] In the second heat transfer medium circuit **1***b*, the heat transfer medium having undergone heat exchange in the oil cooler **461** (and the heat exchanger of PCU **450**) flows through the chiller **610**. The heat supplied to the refrigeration cycle **600** through the chiller **610** is used for heating by the heater core **330** in the high-temperature circuit **300**. Since the heat transfer medium does not flow in LT radiator **410**, heat exchange between the heat transfer medium and the ambient air in LT radiator **410** is not performed.

[0084] In the second heat transfer medium circuit **1***b*, similarly to the first heat transfer medium circuit **1***a*, the heat transfer medium does not flow through each of the flow path **30**, the flow path **40**, the flow path **50**, and the flow path **60**. In the second heat transfer medium circuit **1***b*, the heat transfer medium does not flow through the flow path **490**, similarly to the first heat transfer medium circuit **1***a*.

Third Heat Transfer Medium Circuit

[0085] FIG. **5** is a diagram illustrating a third heat transfer medium circuit **1***c* of the thermal management circuit **1**. The third heat transfer medium circuit **1***c* is a circuit for passing water to the respective devices of the thermal management circuit **1**. Thus, it is possible to eliminate the liquid accumulation and the like.

[0086] The internal flow path **111** of the nine-way valve **100** connects the outer section **122** and the outer section **124**. The internal flow path **112** connects the outer section **123** and the outer section **125**. The internal flow path **113** connects the outer section **121** and the outer section **127**. The internal flow path **114** connects the outer section **120** and the outer section **126**.

[0087] In the five-way valve **200**, the opening **210** overlaps the lower compartment that communicates with each of the port P1 and P2. Thus, the port P**5** is in communication with the port P**1** of P**4** and P2 from the port P**1**.

[0088] Thus, the heat transfer medium circulates through the first circuit of the five-way valve **200** (port P**5**)-water pump **310**-water cooled condenser **640**-HVH **320**-heater core **330**-five-way valve **200** (port P**2**).

[0089] The heat transfer medium also circulates through the second circuit of the five-way valve **200** (port P**5**)-water pump **310**-water cooled condenser **640**-HVH **320**-HT radiator **340**-five-way valve **200** (port P1).

[0090] The heat transfer medium of the second circuit is branched into the flow path **30** at the branch point **371**. The heat transfer medium branched into the flow path **30** circulates in the third circuit. The third circuit is the circuit of LT radiator **410**-reserve tank **420**-water pump **430**-SPU **440**-PCU **450**-oil cooler **461**-outer section **120**-inner section **114**-outer section **126**-chiller **610**-outer section **125**-inner section **112**-outer section **123**-water pump **520**-battery **510**-outer section **124**-inner section **111**-outer section **122**-flow path **800**-outer section **121**-inner section **113**-outer section **127**-LT radiator **410**.

[0091] The heat transfer medium of the third circuit is branched into the flow path **50** at the branch point **531**, and flows into the high-temperature circuit **300** from the branch point **380**.

[0092] In the third heat transfer medium circuit **1***c*, the heat transfer medium does not flow through each of the flow path **40** and the flow path **60**. In the third heat transfer medium circuit **1***c*, the heat transfer medium does not flow through the flow path **490**.

ECU Control Flowchart

[0093] FIG. **6** is a diagram illustrating an exemplary control flow executed by ECU **2** (FIG. **1**). The control flow illustrated in FIG. **6** may be executed (started) at a predetermined cycle (for example, every minute).

[0094] In S**1**, ECU **2** determines whether the heating flag is ON. The heating flag is ON when the user performs a predetermined operation for requesting heating in HMI **21** (FIG. **1**). If the heating flag is ON (Yes in S**1**), the process proceeds to S**2**. When the heating flag is OFF (No in S**1**), the process ends.

[0095] In S**2**, ECU **2** determines whether the second heat transfer medium circuit **1***b* (FIG. **4**) is formed. For example, ECU **2** may make the above determination from the status of the nine-way valve **100** and the five-way valve **200**. When the second heat transfer medium circuit **1***b* is formed (Yes in S**2**), the process proceeds to S**3**. If the second heat transfer medium circuit **1***b* is not formed (No in S**2**), the process proceeds to S**4**.

[0096] In S**3**, ECU **2** determines whether the detected value of the temperature sensor **451** (FIG. **2**) is less than the ambient air temperature. Specifically, ECU **2** determines whether or not the detected value of the temperature sensor **451** is less than the detected value of the ambient air temperature sensor **22** (FIG. **1**). When the detected value of the temperature sensor **451** is less than the ambient air temperature (Yes in S**3**), the process proceeds to S**5**. When the detected value of the temperature sensor **451** is equal to or higher than the ambient air temperature (No in S**3**), the process proceeds to S**6**. The thresholds in S**3** may be, for example, the ambient air temperature+α (for example, α=3° C.) instead of the ambient air temperature.

[0097] In S**4**, ECU **2** determines whether the detected value of the temperature sensor **411** (FIG. **2**) is less than the ambient air temperature. Specifically, ECU **2** determines whether or not the detected value of the temperature sensor **411** is less than the detected value of the ambient air temperature sensor **22** (FIG. **1**). When the detected value of the temperature sensor **411** is less than the ambient air temperature (Yes in S**4**), the process proceeds to S**5**. When the detected value of the temperature sensor **411** is equal to or higher than the ambient air temperature (No in S**4**), the process proceeds to S**6**.

[0098] In S**5**, ECU **2** controls the five-way valve **200** and the nine-way valve **100** to switch the thermal management circuit **1** to the first heat transfer medium circuit **1***a* (FIG. **3**). When the thermal management circuit **1** is switched to the first heat transfer medium circuit **1***a*, the condition is maintained.

[0099] As a result, a series circuit **900** (FIG. **3**) in which LT radiator **410**, the oil cooler **461**, and the chiller **610** are connected in series is formed. Therefore, the heat transfer medium absorbed in each of LT radiator **410** and the oil cooler **461** flows through the chiller **610**. As a result, the amount of heat transmitted to the refrigeration cycle **600** through the chiller **610** becomes relatively large, and thus the power consumption of the compressor **630** can be reduced. This improves the efficiency of heating by the heater core **330**.

[0100] In S**6**, ECU **2** controls the five-way valve **200** and the nine-way valve **100** to switch the thermal management circuit **1** to the second heat transfer medium circuit **1***b* (FIG. **3**). When the thermal management circuit **1** is switched to the second heat transfer medium circuit **1***b*, the condition is maintained.

[0101] As a result, the series circuit **910** (FIG. **4**) in which the oil cooler **461** and the chiller **610** are connected in series is formed, so that the heat transfer medium absorbed by the oil cooler **461** flows through the chiller **610**. Further, since the heat transfer medium that is flowing through the chiller **610** does not pass through LT radiator **410**, heat is not dissipated to the ambient air in LT radiator **410**. As a consequence, the amount of heat transferred to the refrigeration cycle **600** in the chiller **610** can be suppressed from decreasing due to heat dissipation in LT radiator **410**. Accordingly, it is possible to suppress an increase in power consumption of the compressor **630**. As a result, lowering of the heating efficiency by the heater core **330** is suppressed.

[0102] ECU **2** may also control the five-way valve **200** and the nine-way valve **100** while the heating flag is OFF to switch the thermal management circuit **1** to the third heat transfer medium circuit **1***c* (FIG. **5**).

[0103] As described above, in the present embodiment, when heating is requested, ECU **2** controls the nine-way valve **100** and the five-way valve **200** when the temperature of the heat transfer medium that is flowing through LT radiator **410** is lower than the ambient air temperature. Thus, the first heat transfer medium circuit **1***a* in which the oil cooler **461**, LT radiator **410**, and the chiller **610** are connected is formed. ECU **2** also controls the nine-way valve **100** and the five-way valve **200** when the temperature of the heat transfer medium that is flowing through LT radiator **410** is equal to or higher than the ambient air temperature. Thus, the second heat transfer medium circuit **1***b* in which LT radiator **410** is disconnected from the first heat transfer medium circuit **1***a* is formed. Thus, by forming the first heat transfer medium circuit **1***a*, heat absorbed from the ambient air in LT radiator **410** can be used for heating. Further, by forming the second heat transfer medium circuit **1***b*, it is possible to suppress the heat dissipation from the heat transfer medium to the ambient air in LT radiator **410** from lowering the heating efficiency.

[0104] In the above embodiment, LT radiator **410**, the oil cooler **461**, and the chiller **610** are connected in series so that both LT radiator **410** and the oil cooler **461** contribute to heating, but the present disclosure is not limited thereto. The circuit to which LT radiator **410** and the chiller **610** are connected and the circuit to which the oil cooler **461** and the chiller **610** are connected may be separated and independent of each other.

[0105] In the above embodiment, the heat transfer medium circuit (**1***a*, **1***b*) is switched based on the magnitude relation between the temperature of the heat transfer medium immediately before flowing into LT radiator **410** (flowing through the heat transfer medium inlet of LT radiator **410**) and the ambient air temperature. For example, the heat transfer medium circuit may be switched based on the magnitude relation between the temperature of the heat transfer medium immediately after flowing out of LT radiator **410** (flowing through the heat transfer medium outlet of LT radiator **410**) and the ambient air temperature. In this case, the first heat transfer medium circuit **1***a* may be formed when the temperature of the heat transfer medium that is flowing through the heat transfer medium outlet is lower than the ambient air temperature (or the ambient air temperature+α). In addition, the second heat transfer medium circuit **1***b* may be formed when the temperature of the heat transfer medium that is flowing through the heat transfer medium outlet is equal to or higher than the ambient air temperature (or the ambient air temperature+α).

[0106] In the above-described embodiment, the switching of the heat transfer medium circuit (**1***a*, **1***b*) is performed based on the magnitude relation between the detected value of the temperature sensor **451** and the ambient air temperature when the second heat transfer medium circuit **1***b* is formed. Even when the second heat transfer medium circuit **1***b* is formed, the heat transfer medium circuit may be switched based on the magnitude relation between the detected value of the temperature sensor **411** and the ambient air temperature.

[0107] In the above embodiment, the state of the thermal management circuit **1** is switched by controlling the nine-way valve **100** and the five-way valve **200**, but the present disclosure is not limited thereto. The switching valve may have other configurations (e.g., a multi-way valve other than a five-way valve and a nine-way valve).

[0108] Although the reference-temperature sensors (**411**, **451**) are switched based on whether or not the second heat transfer medium circuit **1***b* is formed, the present disclosure is not limited thereto. For example, LT radiator **410** may be configured to detect whether or not the heat transfer medium is circulating, and the temperature sensor to be referred to may be switched based on the detection result. For example, the determination may be performed based on a circuit pattern corresponding to the states of the nine-way valve **100** and the five-way valve **200**. Further, the determination may be performed based on a detected value of a sensor (for example, a pressure sensor, a flow rate sensor, or the like) provided at a heat transfer medium inlet or the like of LT radiator **410**.

[0109] In the above embodiment, PCU **450** and the drive unit **460** (the oil cooler **461** and the transaxle **462**) are connected in series. PCU **450** and the drive unit **460** may be connected in parallel.

[0110] In the above embodiment, an example in which the detection value of the ambient air temperature sensor **22** is used as the information of the ambient air temperature has been described, but the present disclosure is not limited thereto. For example, information on the ambient air temperature acquired by communication to the Internet or the like may be used.

[0111] Note that the configurations (processes) of the above-described embodiments and the above-described modification examples may be combined with each other.

[0112] The embodiment disclosed herein shall be construed as exemplary and not restrictive in all respects. The scope of the present disclosure is shown by the claims rather than by the above description of the embodiments, and is intended to include all modifications within the meaning and scope equivalent to those of the claims.

What is claimed is:

1. A thermal management system in which a heat transfer medium circulates, the thermal management system comprising:

a radiator;

a drive device that includes a heat exchanger and that is configured to generate a driving force;

a chiller device;

a switching device that switches a flow path of the heat transfer medium; and

a control device that controls the switching device, wherein, when heating is requested, the control device

controls the switching device to establish a first heat transfer medium circuit in which the heat exchanger, the radiator, and the chiller device are connected, when a temperature of the heat transfer medium that is flowing through the radiator is lower than ambient air temperature, and

controls the switching device to establish a second heat transfer medium circuit in which the radiator is isolated from the first heat transfer medium circuit, when the temperature of the heat transfer medium that is flowing through the radiator is no lower than the ambient air temperature.

2. The thermal management system according to claim **1**, wherein:

the first heat transfer medium circuit includes a first series circuit in which the heat exchanger, the radiator, and the chiller device are connected in series; and

the second heat transfer medium circuit includes a second series circuit in which the radiator is isolated from the first series circuit and the heat exchanger and the chiller device are connected in series.

3. The thermal management system according to claim **1**, further comprising a temperature sensor that detects the temperature of the heat transfer medium that is flowing into the radiator, wherein, when heating is requested, the control device

controls the switching device to establish the first heat transfer medium circuit, when a detected value of the temperature sensor is lower than the ambient air temperature, and

controls the switching device to establish the second heat transfer medium circuit, when the detected value of the temperature sensor is no lower than the ambient air temperature.

4. The thermal management system according to claim **1**, wherein:

the switching device includes a nine-way valve and a five-way valve; and

the control device switches between the first heat transfer medium circuit and the second heat transfer medium circuit by controlling each of the nine-way valve and the five-way valve.

5. The thermal management system according to claim **1**, further comprising a temperature sensor that detects the temperature of the heat transfer medium that is flowing through the heat exchanger, wherein the control device controls the switching device to establish the first heat transfer medium circuit when a detected value of the temperature sensor becomes lower than the ambient air temperature in a state in which the second heat transfer medium circuit is established.

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